

The Brain Does Not Have Abs

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A Penn State paper measured a real coupling between abdominal muscles and brain motion in mice. Then the internet decided that abdominal movement “literally cleans the brain.” The journey between those two sentences is where most of what we believe about health comes from — and most of what should worry us about it.

A Wellness Headline Walks Into a Lab

On April 27, 2026, the Drew lab at Penn State published a paper in *Nature Neuroscience* with a careful, technically narrow title: “Brain motion is driven by mechanical coupling with the abdomen” [27]. Using two-photon microscopy and microCT in awake mice, the researchers showed that contractions of the abdominal muscles propagate up through the vertebral venous plexus and produce small, measurable shifts in brain position. They then ran computer simulations suggesting that this brain motion could plausibly drive cerebrospinal fluid through the surrounding tissue. The first author, Patrick Drew, told the university’s press office that “more work is needed to determine how these findings apply to humans” [28].

By May 6, the headline at *ScienceAlert* read: “Movement Triggers a Hidden ‘Brain Cleaning’ Mechanism, Study Shows” [29]. Within days, an aggregator at nbsla.ca led with “Shocking Link Between Movement and Brain Health” [30], and a piece at Hoodline announced that “Tiny Tummy Tenses May Help Flush Dementia Waste” [31]. Somewhere in this chain, a contemplative-fitness genre crystallized around a single sentence: *abdominal movement literally cleans the brain*.

That sentence is wrong, in a specific and instructive way. The mouse paper measured brain *motion* and *simulated* fluid flow. It did not measure waste clearance. It did not work in humans. The most scientifically defensible reading of the underlying evidence — synthesizing the Penn State result with two decades of glymphatic research, MRI quantifications of CSF flow, and randomized clinical trials of breathing protocols — is that abdominal motion does measurably modulate cerebrospinal fluid in human beings, but it is a small contributor sitting alongside much larger ones. The phrase “literally cleans the brain,” taken at face value, is not what the science shows. It is the rhetorical residue of a careful paper passing through five layers of paraphrase.

This kind of slippage matters now because the wellness market that turns mouse data into morning routines has scaled. The Penn State paper went from preprint to TikTok talking point in under two weeks. Practitioners are already selling “lymphatic ab pumps” and “glymphatic breathwork” courses citing it. The question is not whether breathing or movement matters for brain health — both clearly do. The question is whether the specific causal story being told to consumers — *contract your abs, flush your brain* — survives contact with the actual evidence.

It does not.

What the Mouse Paper Actually Showed

The Drew lab’s contribution is real and worth understanding precisely. They placed awake mice under a two-photon microscope and tracked brain motion at the cellular level while monitoring abdominal muscle activity. When the mice contracted their abdominal muscles — during locomotion, in response to small applied pressures, or spontaneously — the brain shifted on a measurable scale, and returned to baseline when the contraction relaxed. They then traced the anatomical pathway: the contraction raises pressure inside the abdominal cavity, which pushes blood up through the vertebral venous plexus, which displaces tissue in the cranial vault, which moves the brain.

That is a hydraulic pump. It is novel. It identifies a previously underappreciated mechanical pathway connecting abdominal motion to the cranial fluid environment.

What the paper did not do, and what its corresponding author explicitly cautioned against extrapolating, is measure waste clearance. The simulations of CSF flow are computational predictions, not in vivo measurements. The institutional press release at Penn State preserved this distinction; it said that body movement “may help to cycle cerebrospinal fluid” [28]. *May. Help. Cycle*. Three hedges, all dropped by the second link in the popular-press chain.

The paper also showed effects in mice, on time scales of seconds, with muscle contractions in animals that weigh thirty grams and breathe roughly three times faster than humans. None of that automatically scales. Mice have different vertebral venous anatomy, different intracranial compliance, and different glymphatic kinetics. A mouse-to-human translation step would normally take years of follow-up work; on social media, it took less than a week.

Treating the Penn State paper as the keystone of a “literally cleans the brain” claim is a category mistake. The keystone of a *cleaning* claim is direct measurement of waste removal. That is not what this paper provides.

The Quantitative Receipts

A more useful way to evaluate the proposition is to ask what the human imaging literature already says about how much breathing — the most relevant kind of abdominal movement for everyday life — contributes to cerebrospinal fluid dynamics. Three decades of MRI studies have given us numbers.

In 2015, Steffi Dreha-Kulaczewski and colleagues at Göttingen used real-time MRI in healthy humans and reported that “inspiration is the major regulator of human CSF flow” [1]. The line is striking and has been quoted constantly. It also obscures, on its own, what the field’s quantitative work has clarified since.

In 2017, Yildiz and colleagues used real-time phase-contrast MRI to measure the ratio of respiratory to cardiac contributions to CSF velocity at the upper cervical spine in twenty healthy volunteers [2]. During natural breathing, respiration produced about 14 percent of the velocity that cardiac pulsation produced. Even during deliberate deep breathing, respiration only reached about 40 percent of the cardiac contribution. By this measure, breathing is a real and detectable driver of CSF motion, and it is sub-dominant to the heart.

A 2019 computational study based on long-term invasive intracranial-pressure measurements complicated this picture in a useful way: cardiac pulsation produced larger pressure swings, but respiration produced about five times the *volume* of CSF flow, because slower oscillations move more total fluid even with smaller

pressure peaks [4]. Pressure dominance and flow dominance need not coincide. Respiration is genuinely consequential for net flow; cardiac pulsation is genuinely consequential for the perivascular pumping that drives glymphatic exchange. Both of these things can be true.

Then in 2025, the Yildiz group at Penn used a slow-flow-sensitized phase-contrast MRI technique to address one of the field's lingering criticisms: maybe respiration just sloshes CSF back and forth without producing net directional flow. They reported that respiration enhances *net* CSF flow in awake humans, not merely oscillation [8]. That is the evidence most directly favorable to the proposition's mechanism, and it deserves the weight it has received.

What no one in the imaging literature has shown is that breathing is the *dominant* driver of *clearance* — the actual removal of metabolic waste from brain tissue. That last word is where the proposition lives, and it is also where the evidence weakens fastest.

What Cleaning the Brain Actually Means

The brain does, in fact, get cleaned. The discovery of the glymphatic system by Maiken Nedergaard's lab at Rochester in 2012 [10] established that perivascular spaces along arteries serve as channels through which cerebrospinal fluid percolates into and out of brain tissue, carrying with it metabolic waste — including the amyloid- β fragments implicated in Alzheimer's disease. This system is not the same as CSF flow in the ventricles or around the spinal cord. It is the parenchymal exchange that actually removes interstitial waste.

What drives glymphatic clearance? The Nedergaard lab and its collaborators have spent the last fifteen years answering that question, and the answer is not abdominal motion.

In 2013, Lulu Xie and colleagues showed that during slow-wave sleep the extracellular space in mouse brains expands by roughly sixty percent, and glymphatic clearance increases nearly tenfold relative to wakefulness [6]. Sleep is the dominant condition under which the brain cleans itself. In 2018, Humberto Mestre and colleagues used particle tracking to show that within perivascular spaces, the pumping force is *arterial pulsation* — the wall-motion of blood vessels driven by the heartbeat [5]. And in 2025, Natalie Hauglund and colleagues identified the mechanism that ties these together: norepinephrine-mediated slow vasomotion during NREM sleep, oscillating at about 0.05 hertz, synchronizes blood volume and CSF flow in a way that strongly predicts how much waste gets cleared [7]. The locus coeruleus, releasing norepinephrine in slow waves while you sleep, drives most of the cleaning.

None of these findings depend on, or even mention, abdominal movement.

This is the central asymmetry the proposition's popular framing erases. There is good evidence that breathing modulates CSF flow, including net flow. There is good evidence that the brain has a clearance system. There is much weaker evidence, and in fact some quite specific contrary evidence, that breathing or abdominal motion is a *substantive* driver of *clearance* in healthy adults under normal conditions. The proposition collapses three different statements — *modulates fluid*, *cleans tissue*, *drives the dominant pathway* — into one declarative sentence. The first is supported, the second is conditionally supported with a different primary mechanism, and the third is contradicted.

The Trial That Got Closer to a Real Test

One human study has come closer than most to mechanistically isolating the effect, and the result is worth dwelling on because it does not move the needle in the direction the wellness genre wants.

In 2023, Jungwon Min and colleagues at Penn and elsewhere ran a randomized controlled trial with 108 healthy adults [12]. Participants were assigned to either four weeks of slow-paced breathing with heart-rate-variability biofeedback, or to a control protocol that decreased rather than increased their HRV oscillations. Both groups breathed for twenty to forty minutes daily. The slow-breathing group showed substantial reductions in plasma amyloid- β 40 and amyloid- β 42 — the very biomarkers most relevant to Alzheimer's-style accumulation. The effect size was large and held in younger and older participants.

That looks, at first, like dispositive support for the proposition. Slow breathing reduced amyloid in human blood, in a randomized trial.

But the authors propose a specific mechanism, and it is not abdominal-CSF coupling. They argue that slow-paced breathing stimulates the vagus nerve, which alters autonomic balance, which counteracts noradrenergic signaling that affects amyloid production and clearance. The vagal pathway runs through the brainstem and the autonomic nervous system. It does not run through the vertebral venous plexus or the glymphatic perivascular spaces. Their best causal story is that breathing affects the brain *via the autonomic nervous system*, not via direct fluid mechanics.

A 2018 framework paper by Roderik Gerritsen and Guido Band — *Breath of Life: The Respiratory Vagal Stimulation Model of Contemplative Activity* — laid out the theoretical basis for this in detail [18]. Slow, deep breathing entrains parasympathetic activation, modulates prefrontal-amygdala circuits, and produces measurable cognitive and emotional benefits. None of these effects require abdominal motion to clean the brain in any literal CSF-clearance sense.

So the Min RCT does what the proposition's defenders most want — finds a real biomarker effect of breathing on brain biochemistry — and simultaneously points to the wrong mechanism. The breathing helped. The cleaning, if you want to call it that, came through nerves and not through plumbing.

The Intellectually Honest Reading

Pull this back to the level of what an evidence-following observer should believe.

The most defensible synthesis is this: respiratory and abdominal motion *do* modulate cerebrospinal fluid flow in humans. The mechanism is real, named in the primary literature, and now plausibly extended by the Penn State 2026 hydraulic-pump finding. In specific subpopulations — people with sleep-disordered breathing, older adults whose other clearance mechanisms are degraded, trained breath practitioners with unusually large diaphragmatic excursions — the relative contribution may be larger than at the population mean. CPAP treatment in animal models increases CSF flow and glymphatic transport [14], and untreated obstructive sleep apnea is associated with both impaired glymphatic drainage and cognitive dysfunction in humans [15]. The mechanism is therefore most consequential precisely where breathing is most disrupted.

But at the population mean, in a healthy adult, the abdominal-respiratory contribution to brain clearance is small relative to the dominant drivers — sleep slow-wave activity, norepinephrine-mediated vasomotion during NREM, and arterial pulsation. The Yildiz 2017 ratio of fourteen percent during natural breathing is the right anchor; the Hauglund 2025 result that NE-driven slow vasomotion during sleep is the strongest predictor of clearance is the right ceiling on the alternative. The Penn State 2026 paper increases the credibility of the mechanism without raising the magnitude estimate, because the magnitude question can only be answered by direct human clearance measurements that no one has performed.

Read the proposition literally — *abdominal movement literally cleans the brain* — and the evidence weakly refutes it. Not because the mechanism is fictional, but because the framing overstates how much of the cleaning the named mechanism does. The most likely state of the world is that breathing is a real, sub-dominant

contributor to a fluid system whose dominant cleaning happens at night, when you are not contracting your abdominal muscles and not breathing deeply, while a slow chemical oscillation in your brainstem drags norepinephrine across your vasculature and the spaces between your blood vessels do the actual work.

Sleep, in other words, is doing more for your brain than your abs.

Where the Other Frames Land, and Why

The evidence does not fully resolve the dispute between worldviews that take this kind of question seriously. Three other frames produce visibly different answers, and the way they diverge from the empirical reading is itself diagnostic.

The Wellness-Amplification Reading

A reader sympathetic to the popular-science synthesis — the ScienceAlert headline writer, the breathwork instructor, the wellness podcaster — looks at this evidence base and sees a strong “yes.” A *Nature Neuroscience* paper. A *Nature Communications* paper showing net flow. A randomized trial. Three reviews framing controlled breathing as a glymphatic lever. From inside that frame, the hard-edged “literally” in the proposition reads as accurate.

What that frame does not register is the magnitude question. To say “literally cleans the brain” is to make an implicit claim about how much of the cleaning the named mechanism is doing. The wellness reading treats *mechanism exists* and *mechanism dominates* as effectively the same. They are not. A drug that is one-tenth as effective as the standard of care can still have a real mechanism; calling it the cure mistakes existence for adequacy. The Penn State 2026 paper establishes a real coupling and is consistent with breathing being a small contributor; nothing in it adjudicates magnitude. A sympathetic reader who treats “real” as equivalent to “primary” will arrive at the proposition; an empirical reader who tracks effect-size hierarchy will not.

The wellness reading is also structurally vulnerable to amplification chains. When the original paper’s caveats — *may, might, in mice, simulated, more work needed* — drop out at each retelling, a reader who consumes the genre’s outputs encounters only the strongest forms. By the time the claim reaches “tiny tummy tenses flush dementia waste” [31], every hedge has evaporated, and the reader has no easy way to recover them without going to the primary source. Few do.

The Embodied-Integrative Reading

A different kind of skeptic might object to the question’s framing entirely. From a contemplative-traditions perspective — yoga, qigong, somatic therapy, integrative medicine — the analytical separation of “abdominal movement” from “brain cleaning” is itself the error. Bodies are integrated systems; lived practice across centuries reports breath-mediated cognitive refreshment in dozens of cultures; insisting on a fluid-mechanical mechanism before granting the claim is precisely the reductionist move that misses the phenomenon.

There is something to this. Contemplative traditions did encode practices that produce measurable benefits, and Western neurophysiology is still catching up to which mechanisms are doing the work. The Min trial’s vagal pathway is, in some ways, exactly the kind of finding integrative readers have long anticipated.

But this frame, taken as evidence-evaluation guidance, has a particular weakness: it treats the question’s literal terms as expendable. If “cleans the brain” can mean either CSF clearance or cognitive refreshment depending on convenience, the proposition becomes hard to evaluate at all. The proposition’s defenders cannot have it both ways — claiming a literal physiological mechanism when the popular-science citation chain demands it, then retreating to metaphor when the magnitude evidence pushes back. The intellectually honest version of the integrative position would concede that the literal CSF-clearance claim is overstated while still insisting that breathing-rich practices are valuable for reasons the strict empirical reading captures imperfectly. That is a defensible posture; it is not the posture of the proposition.

The Reductionist-Skeptical Reading

At the opposite pole, a reductionist-skeptical reader looks at the same evidence and sees nothing meaningful. Cardiac pulsation dominates pressure dynamics. Sleep dominates clearance. Arterial pulsation drives the perivascular pumping. Respiration is a fourteen-percent modulator. Most of the human studies are small, the effect sizes are modest, and the only real RCT pointed to a non-CSF mechanism. From this seat, the proposition is essentially negligible — abdominal motion is functionally irrelevant to brain cleaning at the population mean.

The reductionist reading has its own structural blind spot: it treats sub-dominant contributions as zero, and it dismisses subpopulation effects (sleep apnea, aging, trained breath practitioners) as base-rate-irrelevant. A fourteen-percent effect is not nothing; for an older adult with degraded slow-wave sleep, it might be a meaningful fraction of remaining clearance capacity. The reductionist frame is methodologically conservative in a way that minimizes false positives at the cost of missing genuine, distributionally important effects.

The empirical baseline sits between the wellness reading and the reductionist reading — closer to the latter on the literal proposition, closer to the former on the underlying mechanism. The mechanism is real. The proposition’s framing of it is wrong.

What This Pattern Tells Us About Scientific Communication

There is a recurring shape to these mismatches between primary research and popular claims, and the Penn State amplification chain is a textbook example. A careful study reports a measured effect plus a simulation. The institution’s press office hedges appropriately. A first-tier popular-science outlet promotes one of the simulated results into a definite headline. Aggregator sites strip the remaining hedges. By the time the claim reaches the algorithmic feed of a wellness audience, the original mouse-mechanics paper has become a behavioral prescription.

This is not a flaw of any individual writer or outlet; it is a structural property of the citation chain. Each step has its own incentive to flatten ambiguity, and the cumulative effect of small flattening at each step is large. The 2019 paper “Breathing Can Affect the Cleansing of the Brain” at *RealClearScience* [22] illustrates the same dynamic for the Dreha-Kulaczewski lineage; the 2025 *Smithsonian* piece “How Does Breathing Affect Your Brain?” [21] is the more recent version. Across both decades and both research lines, the pattern is the same: real mechanism, amplified into a behavioral declaration.

A useful tell: when a claim says *literally* — *literally cleans the brain, literally rewires your nervous system, literally turns on longevity genes* — the word is almost always doing rhetorical work it cannot afford. *Literal* is a high evidentiary bar. It is reserved for cases where the named mechanism is both established *and* sufficient, where the magnitude is large enough that no honest qualifier is needed. The presence of *literally* in a popular-science headline is, in this genre, a reliable indicator that the qualifier *literally* is being used to suppress the qualifier *primarily*. Andrew Huberman, who hosts one of the most consumed popular-science podcasts about the brain and is generally favorable to integrative practices, declines to make the claim that breathwork is the primary glymphatic lever; his account emphasizes side sleeping, sleep depth, and exercise instead [23]. The careful end of the popular-science distribution does not endorse the proposition.

What to Actually Do About Your Brain

If you have read this far, you may reasonably ask what to do with it. The practical advice is less exotic than the headline genre wants.

First: protect sleep. The single largest evidence-based lever for brain clearance is slow-wave sleep, which is when the norepinephrine-mediated vasomotion that drives glymphatic clearance reaches its peak [7]. Sleep duration, sleep depth, and sleep regularity dominate everything else in the magnitude hierarchy. Side sleeping appears to be modestly better than supine for glymphatic flow, on the basis of animal evidence and inference. If you have to choose one habit to prioritize, this is it.

Second: treat sleep-disordered breathing if you have it. Roughly a quarter of US adults have at least mild obstructive sleep apnea. Untreated OSA is associated with impaired glymphatic drainage and measurable cognitive consequences [15]; CPAP increases CSF flow and glymphatic transport in animal models and is the only intervention in this evidence base where mechanically restoring breathing has a clear, large effect on clearance [14]. The proposition's mechanism is most relevant precisely where breathing is most disrupted, and the actionable form of that insight is screening and treatment, not breathwork apps.

Third: exercise. A 2025 *Nature Communications* paper from the He lab found that twelve weeks of regular exercise enhanced putative glymphatic and meningeal lymphatic flow in humans [16]. A 2017 mouse study from the same group showed similar effects on amyloid- β clearance [17]. Exercise has many parallel mechanisms — cardiovascular conditioning, autonomic regulation, BDNF signaling, sleep improvement — and the abdominal-motion component is one of several. The benefit is real; the specific attribution to abdominal mechanics is not the point.

Fourth: take breathwork seriously, but for the right reasons. The Min RCT is real evidence that slow-paced breathing reduces amyloid biomarkers in healthy adults [12]. The probable mechanism is autonomic, not direct fluid mechanics. The Wu 2024 trial in mild Alzheimer's patients showed that daily deep breathing slowed cognitive decline over 48 weeks [13]; the effect could route through the autonomic system, the CSF system, or both, and current designs cannot separate them. The right framing is: breathing practices have evidence-supported benefits whose mechanism is contested, and the contestation does not undermine the benefit. What it undermines is the specific claim that the benefit comes from cleaning your brain by contracting your abs.

Fifth: when you encounter a claim that uses the word *literally* about a biological mechanism, find the primary citation and read three sentences. The institutional press release is usually a much better filter than the aggregator. The corresponding author's quoted hedge — “more work is needed to determine how these findings apply to humans” — is the part that almost always survives the scientist and almost never survives the social feed. Read it.

The Limits of a Hydraulic Brain

The most useful thing the Penn State 2026 paper offers is a reminder that the brain is, in part, a hydraulic system, and that the body that surrounds it is part of the relevant fluid environment. That is genuinely novel framing for fields that often treat the cranial cavity as a closed compartment. The vertebral venous plexus is real; the abdominal-cranial coupling is real; the simulation suggests CSF consequences worth investigating in humans.

What the paper does not offer, and what no current evidence base offers, is a license for the sentence the wellness genre wants to print. *Abdominal movement literally cleans the brain* is the kind of claim that can only be honestly made on the basis of three sequential findings: that abdominal motion drives CSF flow at substantial magnitudes; that the resulting CSF flow drives glymphatic clearance; and that the clearance is large enough relative to alternative mechanisms to call it cleaning rather than modulation. The first is partially supported; the second is conditionally supported with a different primary driver; the third is not supported. A chain whose middle link is fragile and whose third link is inverted should not be pulled tight.

For now, the best summary is the one Drew gave his university's press office. Body movement may help cycle cerebrospinal fluid. Sleep does the cleaning. The mechanism is interesting; the framing is overstated; the practical implications point to the same boring, evidence-dense advice the field has been giving for a decade. Sleep enough. Treat your apnea. Move your body. Breathe slowly because it is good for your nervous system, not because it is mopping out your skull.

The brain does not have abs. It has a sleeping schedule and a chemistry, and those are what keep it clean.

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